311-211 Endocrine system and metabolism

ข้อสอบ MCQ

1.ข้อสอบมีทั้งหมด 38 ข้อ (ข้อสอบจริงมีจำนวน 68 ข้อ)

2.นักศึกษามีเวลาทำข้อสอบ 120 นาที

3.ข้อสอบจริงเป็นภาษาอังกฤษทั้งหมด

หมายเหตุ : ข้อสอบฉบับนี้เกิดจากความร่วมมือของคนในรุ่นและฝ่ายวิชาการ MED รุ่น 49

1. Which of the following is an effective inhibitor of osteoclast differentiation, activation, and survival?

a. RUNX2
b. RANKL
c. OPG
d. BMP-2
e. Alkaline phosphatase

2. Which of the following is the chemical signal of catecholamine secretion from adrenal medulla?

a. Serotonin
b. Somatostatin
c. Acetylcholine
d. Norepinephrine
e. Vasoactive

3. 30-year-old patient was found abnormal overnight blood values. Genetic testing shows the partial loss function mutation of GLUT-2 gene. Which of the following is found in serum of the patient?

a. high insulin
b. low cortisol
c. elevated LDL-c
d. elevated glucose
e. elevated liver enzymes

4. Which of the following enzyme linked in signal transduction system activated with prolactin receptor?

a. Adenylyl cyclase
b. Guanyl cyclase
c. IPC
d. IPA
e. JAK2-STAT Pathway

5. Which of the following mechanisms involved in glucose homeostasis during fed state?

a. Increased glucose uptake in brain tissue
a. Increased hepatic glucose production
b. Increased hepatic glycogenolysis
c. Increased urine excretion of glucose
d. Increased glucose uptake in muscle tissue

6. กิน niacin คันและเป็นผื่นแดง เกิดจากสารใด

a. prostaglandin
b. histamine
c. acetylcholine
d. p substrate

7. A 55-year-old man กิน thiazide diuretics เพื่อลด hypertension ต่อมาเกิด constipation, fatigue, plasma Ca2+ 12.5 mg/dL โจทย์ถามว่า Thiazide diuretics มี MOA อย่างไร

a. ลด TRPV6 activity
b. ลด intracellular Na+
c. ลด Na+ - K+ - 2Cl- cotransporter activity
d. เพิ่ม Na+ - Cl- cotransporter activity
e. เพิ่ม lla Na+/Pi cotransporter expression

8. เป้าหมายของการวินิจฉัยโรคกระดูกพรุน

a. To predict bone density
b. To predict bone quality
c. To predict fracture risk
d. To predict for survival

9. Which of the following lipoprotein particles providing energy to tissue?

a. VLDL
b. LDL
c. Chylomicron
d. IDL
e. HDL

10. A 58-year-old man had a thyroid cancer removal. Six months later, he had a regular medical check-up. The doctor mentioned that he had an abnormal calcium level (2.0 1.5 mmol/L). Which of the following signs/symptoms might be observed?

a. Dehydration
b. Tetany
c. Osteoporosis
d. Kidney stones
e. Constipation

11. A 38-year-old woman with type 2 diabetics developed severe lactic acidosis after receiving anti-diabetic drugs. Which of the following drug is the most likely cause?

a. Metformin
b. Acarbose
c. Repaglinide
d. Regular insulin
e. Pioglitazone

12. Patient take prednisolone with other drug and then having nausea and weakness. What is the following mechanism of prednisolone cause this symptom?

a. induce CYP3A4
b. inhibit CYP3A4
c. induce 11-BHSD1
d. inhibit 11-BHSD

13. A patient with incomplete mutations of the starD1 gene, show low level of STAR protein expression. This mutation lead to elevated ACTH, adrenal hypertrophy, and engorgement of the adrenal cell with lipid. Which of the following is the symptoms of the patient?

a. salt waste
b. Hypokalemia
c. Hypertension
d. Hyperglycemia
e. Metabolic alkalosis

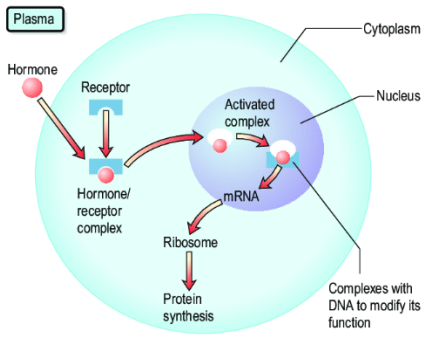
14. A 50-year-old man has received lipid lowering drug which binds to PPAR-α. Which of the following enzyme in lipid metabolism will be increased?

a. PCSK9
b. Lipoprotein lipase
c. HMG-CoA reductase
d. hormone-sensitive lipase
e. diacylglycerol acyltransferase

15. Which of the following mechanisms involved in normalizing plasma glucose in subjects with insulin resistance?

a. Decrease appetite
b. Decreased intestinal glucose reabsorption
c. Increased urinary glucose excretion
d. Increased insulin secretion
e. Increased glucose uptake in brain tissue

16. From this following picture. What is this hormone?



a. Aldosterone
b. ADH
c. Epinephrine
d. Thyroid
e. Parathyroid

17. เด็กชายอายุ 10 วัน น้ำหนักลด ปากแห้ง skin pigmentation โจทย์ถาม gross adrenal

a. Normal size with dark pigmentation
b. Atrophy both gland
c. One normal, one atrophy
d. Diffuse
e. Multinodular

18. แพทย์วินิจฉัยว่าเป็น autosomal recessive disorder เนื่องมาจาก mutation of CYP27B1
Which following is contribute of these signs/symptoms?

a. เพิ่ม calcitonin
b. เพิ่ม TRPV6
c. เพิ่ม Na-Pi 2B
d. ลด 17 hydroxylase
e. ลด PTH

19. 40 years old female มาห้องฉุกเฉิน ด้วยอาการ headache.

Physical examination: heart rate 80 bpm blood pressure 170/110

Lab: serum Na+ 135 mmol/dL

serum K+ 2.5 mmol/dL

serum morning cortisol 11.7 ng/dL

plasma aldosterone 160 ng/dL

plasma renin activity 0.05 ng/dL/hour

What is the most diagnosis?

a. Primary hyperaldosteronism
b. Secondary hyperaldosteronism
c. Primary adrenal insufficiency
d. Secondary adrenal insufficiency
e. Congenital adrenal hyperplasia

20. Which of the following excess cortisol cause in Cushing patient?

a. anemia
b. hypoglycemia
c. hyperkalemia
d. hypercalcemia
e. truncal obesity

21. A 35-year-old female presented with a 6-month history of an 8 kg body weight gain and gradual rounding of the face. Serial examinations revealed a loss of circadian rhythm of cortisol, elevated urinary free cortisol level and undetectable plasma ACTH level (<5 pg./mL). What is the most likely cause in this patient?

a. Carcinoma of bronchus
b. Adrenal adenoma
c. Pituitary tumor
d. Carcinoid tumor
e. Small cell lung cancer

22. Which of followings cause 'escape phenomenon'?

a. Iodide
b. MMI
c. PTU
d. Iodide radioactive
e. propranolol alone

23. Pars nervosa มีผลต่อการทำงานกระบวนการอะไรในร่างกาย

a. ควบคุมการเพิ่มลดของไทรอยด์
b. ควบคุมการ accumulation ของน้ำในร่างกาย
c. ควบคุมปริมาณแคลเซียม
d. ควบคุมการพัฒนาของไข่

24. A 32-year-old woman has symptoms of fatigue, weakness, atrophy, and postural hypotension for 1 month. She hasn't had a period since giving birth to her baby 18 months ago.
Lab results showed: hyponatremia, hypoglycemia, decreased ACTH and cortisol levels. Which of the following is the cause of her symptoms?

a. Sheehan syndrome
b. Waterhouse-Friedrichsen syndrome
c. Addison's disease
d. Conn syndrome
e. Langerhans Cell histiocytosis

25. ให้ total cholesterol = 280 mg/dL, TG =150 mg/dL, HDL-c = 50 mg/dL ถามว่า LDL-c เท่าไหร่
ตอบ

26. A 47-year-old man has lump at neck for 2 months. PE: rubbery-firm nodule in the right thyroid lobe. The cytologic smear from fine needle aspiration reveals groups of follicular having nuclear inclusion and nuclear groove.

a. Follicular carcinoma
b. Nodular goiter

c. Subacute thyroiditis
d. Papillary carcinoma
e. Hashimoto's thyroiditis

27. A 45-year-old male with type 2 diabetes has been prescribed sulfonylureas for blood glucose control. Which of the following choices is the effect of this drug on pancreatic ß cell?

a. ATP-gated K channel is activated
b. Insulin granules are secreted by voltage-gated channel.
c. Voltage-dependent Ca2+ channel is inhibited
d. Decreased intracellular concentration of Ca2+
e. Pancreatic ß cell membrane depolarization

28. คนไข้มี lateral visual defect แต่ residual vision 20/20 และมีอาการคางยื่น มือเท้าใหญ่ จะตรวจพบค่าใดที่มีปริมาณที่สูง

a. ADH
b. GH
c. TSH
d. Prolactin
e. ACTH

29. 55-year-old patient experiences muscle pain & weakness after long term use of dexamethasone. What is the most likely mechanism explaining her symptoms.

a. Decrease glucose uptake
b. Decrease Co-Q10 synthesis
c. Decrease protein synthesis
d. Decrease glycogen synthesis
e. Decrease calcium absorption

30. อายุ 70 ปี กิน colchicine กับ simvastatin เพื่อรักษาโรค gout และ hyperlipidemia ตามลำดับ what parameters could be found in this patient?

a. serum LDL-c
b. serum uric acid

c. creatine kinase
d. plasma glucose concentration
e. plasma colchicine concentration

31. Which of the following is a primary and direct stimulus for glucagon secretion during the fasting state?

a. low insulin
b. high glucose
c. high free fatty acid
d. somatostatin
e. cholinergic innervation

32. Patient with IGF-1 deficiency was treated with mecasermin. The most common ADR?

a. Hypoglycemia
b. Hypokalemia
c. Capel tunnel syndrome
d. Retinopathy
e. Hypotension

33. MOA of anti-thyroid drugs ข้อใดถูกต้อง

a. PTU: peripheral conversion of T4
b. RAI: oxidation
c. MMI: thyroglobulin proteolysis
d. Iodine: coupling
e. Iodine: trapping

34. Which is the following hormone would be increased in patients with damaged pituitary stalk?

a. Prolactin
b. Growth hormone
c. Follicle-stimulating hormone (FSH)
d. Thyroid-stimulating hormone (TSH)
e. Adrenocorticotropic hormone (ACTH)

35. A 65-year-old man presented with low trauma fractures of spines. PH: He had nasopharynx cancer since 35 y/o with head and neck radiation. PE: V/S P 48 /min regular, BP 140/100 mmHg

Physical examination: Puffy face and eyelids, dry skin
Deep tendon reflex: 2+ with slow relaxation
Small testicular size
Back tenderness at LS spine area

What is pathology of his illness?

a. Hypoglycemia
b. Panhypopituitarism
c. Prolactin deficiency
d. Adrenal insufficiency
e. Growth hormone deficiency

36. What is the diagnosis of a patient who has hyperglycemia from total pancreatectomy and requires insulin injection?

a. Type 1 diabetes
b. Type 2 diabetes
c. Other specific type of diabetes
d. Diabetes of undetermined cause
e. Insulin dependent mellitus

37. Which of the following is a disadvantage of radioactive iodine therapy?

a. High cost
b. Increased risk of thyroid cancer
c. Destroying the nearby organs
d. Increased risk of hypothyroidism
e. Contraindicated in elderly

38. Which of the following lipoprotein particles increased when Apo E is abnormal?

a. IDL
b. DHL
c. LDL
d. VLDL
e. Chylomicron